

Java Exception Handling Lab Report

Program: InterruptedException

1. Aim

To demonstrate handling of InterruptedException in Java when a sleeping thread is interrupted.

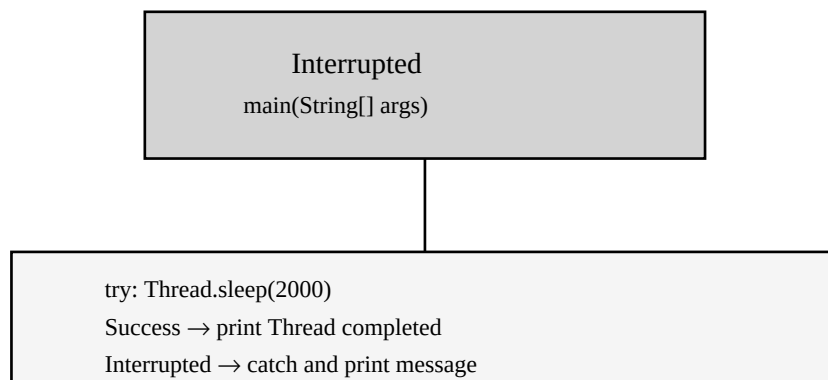
2. Description

InterruptedException is a checked exception that occurs when a thread is interrupted while it is waiting, sleeping, or otherwise paused. In this program, Thread.sleep(2000) pauses the current thread for two seconds. If the thread is interrupted during this period, the catch block displays an appropriate message. Otherwise, the program prints "Thread completed".

3. Tools Used

- Java Development Kit (JDK)
- Java compiler (javac)
- IDE or command prompt
- Thread class from java.lang package

4. UML / Program Structure



5. Algorithm

1. Start the program.
2. Enter the try block.
3. Pause the current thread for 2000 milliseconds using Thread.sleep().
4. If the sleep completes without interruption, display "Thread completed".
5. If the thread is interrupted, catch InterruptedException.
6. Display "Thread was interrupted".
7. Stop the program.

6. Complete Java Source Code

```
package exceptions;
public class Interrupted {
    public static void main(String[] args) {
        try {
```

```
        Thread.sleep(2000);
        System.out.println("Thread completed");
        //Handling interruptions in multithreaded programs
    } catch (InterruptedException e) {
        System.out.println("Thread was interrupted");
    }
}
```

7. Sample Output

Thread completed

Note: The interruption message appears only if the thread is interrupted while sleeping.

8. Result

The program successfully demonstrated handling of `InterruptedException`. The thread sleeps for two seconds and prints a completion message when it is not interrupted; otherwise, the exception is caught and the interruption message is displayed.